When Machine Learning Beats Persistence: A Leakage-Free Streamflow and Drought Benchmark for the Tropical Andes

Richar Andre Vilca-Solorzano, Renato Quispe-Vargas and Dina Maribel Yana-Yucra

Faculty of Statistical and Computer Engineering, Universidad Nacional del Altiplano, Av. Floral 1153, Puno 21001, Peru

Corresponding author: R. A. Vilca-Solorzano, 75521963@est.unap.edu.pe, Tel.: +51 952 190 381
R. Quispe-Vargas, 72535253@est.unap.edu.pe; D. M. Yana-Yucra, maribelbel201314@gmail.com
ORCID: R. A. V.-S. 0009-0003-2385-5263; R. Q.-V. 0009-0005-2367-7852;
D. M. Y.-Y. 0009-0003-6218-2735

Abstract Machine-learning studies of Andean streamflow rarely compare their models with persistence, and their validation designs can leak information from the test period. We built an open benchmark of 19 long-record gauges from the Andean Glacierized Catchment dataset in three regions: the Altiplano, including Puente Isla Cabanillas in the Lake Titicaca basin, the Cordillera Blanca, and the western and southern Andes. Daily discharge was combined with ERA5 and MERRA-2 forcing and GloFAS v4 simulated discharge. Six learners were compared with persistence and climatology at 1-, 3- and 7-day leads under a calendar-continuous, embargoed protocol tuned on three validation blocks. Uncertainty was estimated with block-bootstrap intervals and Diebold–Mariano tests. The learners beat persistence at 16 of 19 gauges at one day and at 17 of 19 at seven days (median skill 0.108). Lower error did not improve drought detection: the median critical success index of drought days was 0.43 for persistence and 0.25 for the learners. A drought threshold calibrated on validation data raised it to 0.40, close to but still below persistence (0.45). Drought synchrony decayed over about 582 km.

Key words: Tropical Andes · Altiplano · Streamflow forecasting · Hydrological drought · Machine learning · Data leakage

1 Introduction

Rivers of the tropical Andes supply irrigation, hydropower and drinking water, but their flow is strongly seasonal and exposed to glacier retreat and drought [24]. On the Altiplano, rivers such as the Coata–Cabanillas feed Lake Titicaca and depend on a short austral-summer wet season [7], and hydrological droughts are projected to become longer and more frequent [24]. Short-range forecasts are therefore useful, but gauges are sparse and records gappy.

Table 1 Data-driven streamflow studies in the Andes. Pers.: persistence reference; CI: uncertainty of skill. Scores as reported by the authors.

Study	Region	Gauges	Step, lead	Method	Pers.	CI	Reported score
Muñoz et al. 2018 [13]	Ecuador	1	4–24 h	RF	no	no	NSE 0.76–0.38
Contreras et al. 2021 [3]	Ecuador	1	4–24 h	RF	no	no	R^2 0.69–0.29
Merizalde et al. 2023 [12]	Ecuador	1	1–11 h	LSTM	no	no	NSE 0.93–0.67
de la Fuente et al. 2019 [6]	Chile	9	hourly, 3 d	LSTM	no	no	NSE ≈0.93–0.99
Núñez et al. 2023 [15]	Chile	1	1–4 months	RF	no	no	R^2 0.85–0.73
Teverovsky et al. 2023 [20]	Argentina	1	1 month	SVR	no[a]	no	NSE up to 0.90
Vásquez-R. et al. 2023 [21]	Peru	1	monthly	ANN	no	no	r^2 0.81
Lujano et al. 2023 [11]	Peru (Altiplano)	1	1 day	k-NN	no	no	NSE 0.979
This study	Peru, Chile	19	1–7 days	6 learners	yes	yes	skill 0.108 at 7 days

[a]Compared with seasonal ARIMA.

Machine learning (ML) is widely used to forecast streamflow [14]. At daily leads, however, discharge is so autocorrelated that persistence, $\hat{Q}_{t+h} = Q_t$, is hard to beat [16]. Reported gains can be inflated by weak references, by shuffled cross-validation that leaks temporal information [2, 22], and by giving the baseline and the learners different information at the time of issue. An earlier version of our own pipeline made the last error, and gave the learners discharge only up to $t-1$.

We measured how much ML adds to persistence for 1- to 7-day streamflow and drought forecasts in the Andes when no information leaks from the test period. The paper makes four contributions: (i) an open benchmark of 19 gauges in three hydroclimatic regions; (ii) a leakage-free protocol checked by automated tests; (iii) skill against persistence with confidence intervals, compared with earlier Andean studies; and (iv) a direct, calibrated evaluation of drought-day detection. Section 2 reviews related work, Sect. 3 describes the data and methods, Sect. 4 presents the results, Sect. 5 discusses them with their limitations, and Sect. 6 concludes.

2 Related Work

Random forests and LSTM networks have been used for hourly runoff forecasting in Ecuadorian Andean catchments [3, 12, 13]. In Chile, LSTM networks supported flood warning in the Maipo basin [6], and random forests predicted monthly flows in the Elqui basin [15]. Support-vector regression was tested for monthly runoff in the Central Andes of Argentina [20]. In Peru, neural networks forecast monthly flows in the Crisnejas basin [21], and k-nearest neighbours forecast daily flows of the Ramis River on the Altiplano [11]. Table 1 summarizes these studies. Seven of the eight used a single target gauge, none compared the models with persistence, and none reported an uncertainty estimate for the skill. For Peruvian agricultural data, blocked cross-validation showed that random folds overstate accuracy [22]. Drought forecasts also need an explicit identification step [19], so we score drought days directly.

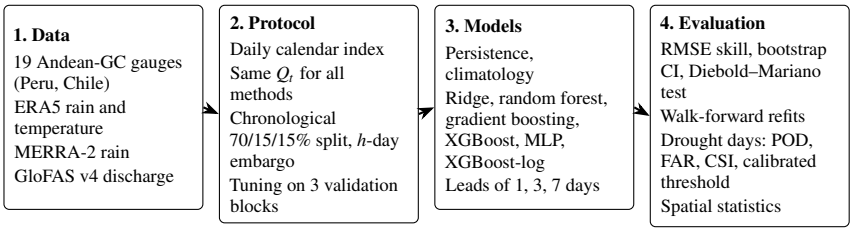


Fig. 1 Workflow of the benchmark. Code, tests and result tables are in the public repository.

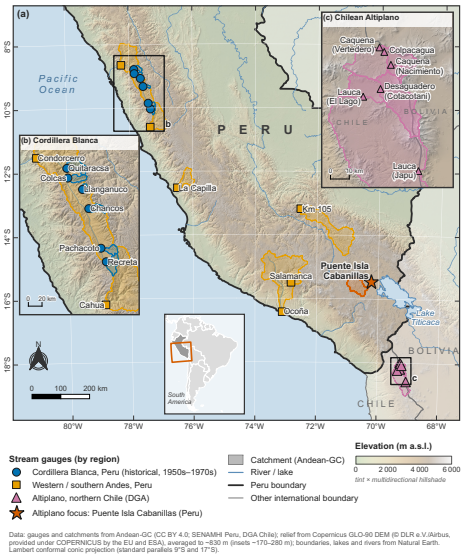


Fig. 2 The 19 gauges and their catchments by region. The focus gauge, Puente Isla Cabanillas, is highlighted.

3 Data and Methods

Data. Figure 1 summarizes the workflow. We used quality-controlled (QC) daily discharge from the Andean-GC dataset [1]. We kept every Peruvian gauge of the National Meteorological and Hydrological Service (SENAMHI) and every Chilean Altiplano gauge of the General Directorate of Water (DGA) with at least 5000 QC days. This gave 19 gauges: 7 on the Altiplano, 6 in the glacier-fed Cordillera Blanca, and 6 in the western and southern Andes (Fig. 2). The focus gauge was Puente Isla Cabanillas (2843 km^2, mean elevation 4471 m), which drains into Lake Titicaca. Daily precipitation and 2-m temperature came from ERA5 [8] at each catchment centroid through Open-Meteo [23]. MERRA-2 precipitation and GloFAS v4 discharge were added in a data-fusion test.

Forecasts and learners. A forecast issued on day t predicts Q_{t+h}, $h \in \{1,3,7\}$ days, from 27 predictors available at t. These were discharge lags (0 to 30 days), 7- to 30-day rolling means and standard deviations, precipitation lags and 7- to 90-day

totals, temperature means, and the day of year. A lag of k days is exactly k calendar days, and samples touching a gap were dropped. The references were persistence, using the same Q_t as the learners, and a day-of-year climatology. The six learners were Ridge regression, random forest, histogram gradient boosting, XGBoost, a two-layer perceptron (MLP), and an XGBoost trained on the change in $\log(1+Q)$ (XGBoost-log).

Protocol. Each record was split chronologically into 70% training, 15% validation and 15% test, with an h-day embargo between blocks. Hyper-parameters and the target form (level or change) were chosen on the mean root-mean-square error (RMSE) over three chronological blocks: the validation block and two earlier blocks of the same length cut from training. The chosen model was refit on training plus validation and scored once on test. Ridge was also run in a three-fold expanding walk-forward at every gauge. Automated tests check calendar offsets, the absence of future information and the embargo.

Scores. We report the Nash–Sutcliffe efficiency (NSE) and the skill $S = 1 - \mathrm{RMSE}_{\mathrm{model}}/\mathrm{RMSE}_{\mathrm{pers}}$. Its 95% confidence intervals (CIs) came from a moving-block bootstrap [9], and equal accuracy was tested with the Diebold–Mariano (DM) test [4]. A day was in drought when Q fell below the monthly 20th percentile of the fitting period [19]. Detection was scored with the probability of detection (POD), false-alarm ratio (FAR) and critical success index (CSI) [18]. Because RMSE-trained models are conservative near the threshold, we also calibrated the drought decision: a forecast signals drought when $\hat{Q} < k\,Q_{20}$, with $k \in [0.8, 1.8]$ chosen to maximize CSI on the validation block. The same calibration was applied to every method, including persistence. Drought synchrony was measured with the correlation of a nonparametric 30-day standardized streamflow index (SSI-30) [5] between gauge pairs, fitted as $\rho(d) = c\,e^{-d/d_0}$ against distance d. Spatial clustering of skill was tested with Moran's I.

4 Results

Focus gauge. At Cabanillas (Table 2; test period 2021-09-22 to 2024-12-24), the validation-selected random forest reached NSE $= 0.912$ at one day, against 0.902 for persistence. This is a skill of 0.054 (95% CI -0.003 to 0.113; DM p 0.048), whereas our earlier, leaky pipeline had found a loss. At seven days, persistence fell to NSE $= 0.461$, and the selected Ridge regression gained 0.115 (95% CI 0.028 to 0.208); its walk-forward folds gave 0.166, 0.116, 0.114. Ridge was better than persistence at all three leads with CIs above zero (Table 2).

All gauges. The learner beat persistence at 16/19, 17/19 and 17/19 gauges at 1, 3 and 7 days, and its CI excluded zero at 11, 12 and 12 gauges (Fig. 3). The median skill grew from 0.051 to 0.080 and 0.108, because persistence degraded faster than the learners. Between one and seven days it rose in all three regions: from 0.060 to 0.103 on the Altiplano, from 0.033 to 0.097 in the Cordillera Blanca and from 0.053 to 0.130 in the western and southern Andes. Ridge, the most often selected

Table 2 Test scores at Puente Isla Cabanillas. Skill is relative to persistence, with 95% CI. †Learner selected on validation. CSI: drought-day detection at seven days.

	1-day lead		7-day lead		
Method	NSE	Skill [95% CI]	NSE	Skill [95% CI]	CSI
Persistence	0.902	–	0.461	–	0.61
Ridge	0.915	0.070 [0.01, 0.11]	0.578	0.115† [0.03, 0.21]	0.47
Random forest	0.912	0.054† [−0.00, 0.11]	0.528	0.064 [−0.02, 0.14]	0.59
Hist. gradient boosting	0.916	0.074 [0.01, 0.14]	0.437	−0.022 [−0.16, 0.12]	0.18
XGBoost	0.915	0.071 [−0.01, 0.13]	0.434	−0.025 [−0.17, 0.13]	0.56
MLP	0.932	0.169 [0.13, 0.20]	0.183	−0.231 [−0.51, −0.03]	0.25
XGBoost-log	0.916	0.077 [0.00, 0.13]	0.502	0.038 [−0.08, 0.18]	0.56

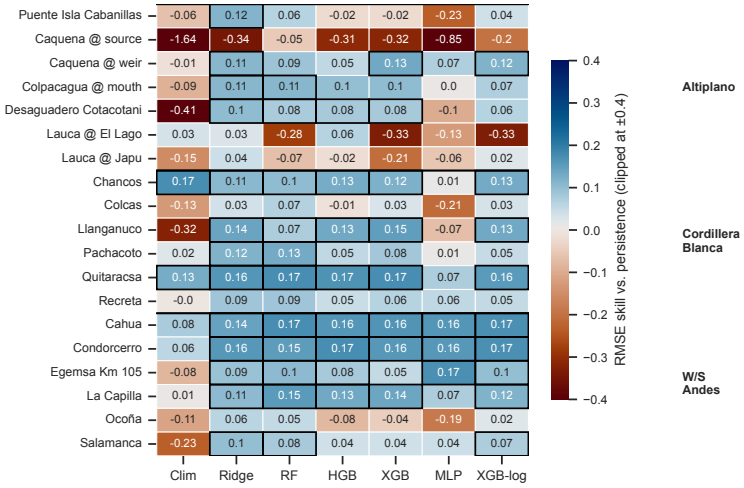


Fig. 3 Seven-day RMSE skill against persistence for every gauge and method. Boxed cells: 95% CI above zero. Gauges are grouped by region (top to bottom: Altiplano, Cordillera Blanca, western and southern Andes).

learner, was skilful at 18/19 gauges at seven days, and 53 of 57 walk-forward folds had positive skill (median 0.103).

Drought detection. Lower RMSE did not give better detection of drought days (Fig. 4). At seven days, the median CSI was 0.43 for persistence and 0.25 for the learner, which beat persistence at only 3/19 gauges. The learners raised fewer false alarms (median FAR 0.38 vs. 0.39) but missed more drought days (POD 0.44 vs. 0.60). Calibration moved the learners towards more warnings (POD 0.68, FAR 0.53) and improved their CSI at 11 gauges, to a median of 0.40. Calibrated persistence reached 0.45 and remained better at 16 gauges.

Spatial statistics. The SSI-30 correlation of 104 gauge pairs decreased with distance (Spearman −0.56, $p < 0.001$), with $d_0 = 582$ km (95% CI 330–1069 km), and was higher within regions than across them (median 0.28 vs. 0.09). Skill showed no spatial clustering (Moran's I =-0.012, $p = 0.540$), and no catchment attribute was significant after Holm correction (strongest: snow fraction, ρ =-0.45, $p = 0.056$).

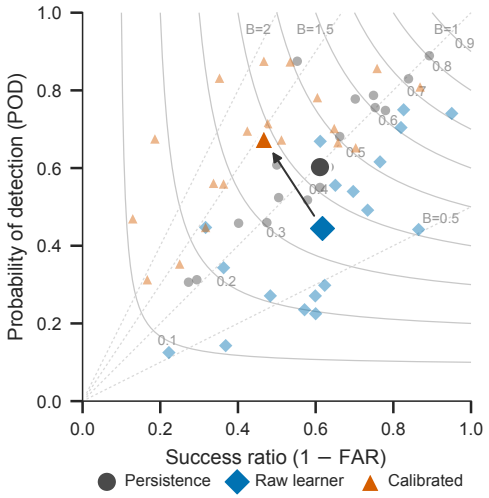


Fig. 4 Performance diagram [17] of seven-day drought-day detection. Small marks: gauges; large marks: medians. Curves: CSI (labelled on the diagonal); dotted lines: frequency bias B. The arrow shows the effect of calibrating the drought threshold.

5 Discussion

The largest change in our conclusions came from the protocol, not from the model. When both received the same information and lags followed the calendar, a one-day loss at Cabanillas became a gain. The gains are modest (5–10% lower RMSE) but consistent, and Ridge was as good as tree ensembles [14].

This helps to read the earlier Andean studies (Table 1). Their scores are high, often NSE above 0.9 at short leads, but none was compared with persistence. At Cabanillas, persistence alone reached NSE $= 0.902$ at one day and the best learner 0.912. For the Ramis River, a k-NN model with only the previous day's discharge reached NSE $= 0.965$, against 0.979 for the best model [11]. In both rivers most of the NSE comes from persistence, so a high NSE is not evidence of skill at daily leads.

Drought detection did not follow RMSE because models trained on squared error smooth the recession limb near the threshold. Calibrating the decision on validation data closed most of this gap, but persistence remained the better drought reference, so drought warnings should be trained and verified on the drought decision itself [19]. Adding MERRA-2 and GloFAS changed the seven-day Ridge skill by a median of only -0.005, and a 0.25° reanalysis resolves the short convective pulses of the Altiplano only in part [7, 10].

Limitations. ERA5 is coarse for steep catchments. The Cordillera Blanca records predate MERRA-2 and GloFAS, so the fusion test covered 7 gauges only. Three selection blocks reduce, but do not remove, the effect of non-stationarity such as glacier retreat. Predictors used observed forcing, not weather forecasts, and Bolivian records were not available.

6 Conclusion

We released an open, leakage-free benchmark for 1- to 7-day streamflow and drought forecasting at 19 Andean gauges. The learners beat persistence at 17/19 gauges at seven days (median skill 0.108). The gains did not carry over to drought-day detection; a calibrated threshold narrowed the gap but did not close it. Future work will add drought-aware training losses, operational weather forecasts and more Altiplano gauges.

Declarations

Data and code availability. Discharge: Andean-GC, doi:10.5281/zenodo.18035801. Code, leakage tests, gauge list, result tables and supplementary figures: `https://github.com/Andre031222/streamflow-drought-altiplano`. **Author contributions.** R. A. Vilca-Solorzano conceived the study, wrote the software and the manuscript; R. Quispe-Vargas validated the results; D. M. Yana-Yucra supervised the study. All authors revised and approved the manuscript. **Competing interests.** None. **Use of AI.** Generative AI was used to improve the language; the authors reviewed the text and take full responsibility for it.

References

1. Aguayo, R., Zekollari, H., van Tiel, M., Bolibar, J., Van Tricht, L., Ayala Ramos, A.I., Ultee, L.: Andean Glacierized Catchment (Andean-GC) dataset. Zenodo dataset (2026). DOI 10.5281/zenodo.18035801
2. Bergmeir, C., Benítez, J.M.: On the use of cross-validation for time series predictor evaluation. Information Sciences **191**, 192–213 (2012). DOI 10.1016/j.ins.2011.12.028
3. Contreras, P., Orellana-Alvear, J., Muñoz, P., Bendix, J., Célleri, R.: Influence of random forest hyperparameterization on short-term runoff forecasting in an Andean mountain catchment. Atmosphere **12**(2), 238 (2021). DOI 10.3390/atmos12020238
4. Diebold, F.X., Mariano, R.S.: Comparing predictive accuracy. Journal of Business & Economic Statistics **13**(3), 253–263 (1995). DOI 10.1080/07350015.1995.10524599
5. Farahmand, A., AghaKouchak, A.: A generalized framework for deriving nonparametric standardized drought indicators. Advances in Water Resources **76**, 140–145 (2015). DOI 10.1016/j.advwatres.2014.11.012
6. de la Fuente, A., Meruane, V., Meruane, C.: Hydrological early warning system based on a deep learning runoff model coupled with a meteorological forecast. Water **11**(9), 1808 (2019). DOI 10.3390/w11091808
7. Garreaud, R., Vuille, M., Clement, A.C.: The climate of the Altiplano: observed current conditions and mechanisms of past changes. Palaeogeography, Palaeoclimatology, Palaeoecology **194**(1–3), 5–22 (2003). DOI 10.1016/S0031-0182(03)00269-4
8. Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L.,

Healy, S., Hogan, R.J., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., de Rosnay, P., Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.N.: The ERA5 global reanalysis. Quarterly Journal of the Royal Meteorological Society **146**(730), 1999–2049 (2020). DOI 10.1002/qj.3803

9. Künsch, H.R.: The jackknife and the bootstrap for general stationary observations. The Annals of Statistics **17**(3), 1217–1241 (1989). DOI 10.1214/aos/1176347265
10. Lujano, E., Diaz, R.D., Lujano, R., Sanchez-Delgado, M., Lujano, A.: Hydrological performance of gridded meteorological products in Peruvian Altiplano basins. RBRH – Revista Brasileira de Recursos Hídricos **30**, e10 (2025). DOI 10.1590/2318-0331.302520240068
11. Lujano, E., Lujano, R., Huamani, J.C., Lujano, A.: Hydrological modeling based on the KNN algorithm: An application for the forecast of daily flows of the Ramis river, Peru. Tecnología y Ciencias del Agua **14**(2), 169–203 (2023). DOI 10.24850/j-tyca-14-02-05
12. Merizalde, M.J., Muñoz, P., Corzo, G., Muñoz, D.F., Samaniego, E., Célleri, R.: Integrating geographic data and the SCS-CN method with LSTM networks for enhanced runoff forecasting in a complex mountain basin. Frontiers in Water **5**, 1233,899 (2023). DOI 10.3389/frwa.2023.1233899
13. Muñoz, P., Orellana-Alvear, J., Willems, P., Célleri, R.: Flash-flood forecasting in an Andean mountain catchment—development of a step-wise methodology based on the random forest algorithm. Water **10**(11), 1519 (2018). DOI 10.3390/w10111519
14. Nearing, G.S., Kratzert, F., Sampson, A.K., Pelissier, C.S., Klotz, D., Frame, J.M., Prieto, C., Gupta, H.V.: What role does hydrological science play in the age of machine learning? Water Resources Research **57**(3), e2020WR028,091 (2021). DOI 10.1029/2020WR028091
15. Núñez, J., Cortés, C.B., Yáñez, M.A.: Explainable artificial intelligence in hydrology: Interpreting black-box snowmelt-driven streamflow predictions in an arid Andean basin of north-central Chile. Water **15**(19), 3369 (2023). DOI 10.3390/w15193369
16. Pappenberger, F., Ramos, M.H., Cloke, H.L., Wetterhall, F., Alfieri, L., Bogner, K., Mueller, A., Salamon, P.: How do I know if my forecasts are better? Using benchmarks in hydrological ensemble prediction. Journal of Hydrology **522**, 697–713 (2015). DOI 10.1016/j.jhydrol.2015.01.024
17. Roebber, P.J.: Visualizing multiple measures of forecast quality. Weather and Forecasting **24**(2), 601–608 (2009). DOI 10.1175/2008WAF2222159.1
18. Schaefer, J.T.: The critical success index as an indicator of warning skill. Weather and Forecasting **5**(4), 570–575 (1990). DOI 10.1175/1520-0434(1990)005<0570:TCSIAA>2.0.CO;2
19. Sutanto, S.J., Van Lanen, H.A.J.: Streamflow drought: implication of drought definitions and its application for drought forecasting. Hydrology and Earth System Sciences **25**(7), 3991–4023 (2021). DOI 10.5194/hess-25-3991-2021
20. Teverovsky Korsic, S.A., Notarnicola, C., Uriburu Quirno, M., Cara, L.: Assessing a data-driven approach for monthly runoff prediction in a mountain basin of the Central Andes of Argentina. Environmental Challenges **10**, 100,680 (2023). DOI 10.1016/j.envc.2023.100680
21. Vásquez-Ramírez, L., Vásquez-Paredes, L.F.: Predicción de caudales mensuales en ríos de cuencas altoandinas con enfoque de redes neuronales artificiales. Caso: río Crisnejas, Perú. Tecnología y Ciencias del Agua **14**(1), 124–199 (2023). DOI 10.24850/j-tyca-14-01-04
22. Vilca Apaza, W., Torres Cruz, F., Carpio Vargas, E.E.: Blocked and hierarchical cross-validation for agricultural data with spatial autocorrelation: Application to Peru's national agricultural survey. In: S. Duarte Correia, M. Lahby (eds.) Green Technologies and Sustainable Intelligence—Volume II: Applications, Case Studies, and Future Challenges, *Springer Proceedings in Physics*, vol. 352, pp. 3–13. Springer Nature Switzerland, Cham (2026). DOI 10.1007/978-3-032-20151-5_1
23. Zippenfenig, P.: Open-Meteo.com Weather API. Zenodo software, version 1.4.0 (2024). DOI 10.5281/zenodo.7970649
24. Zubieta, R., Molina-Carpio, J., Laqui, W., Sulca, J., Ilbay, M.: Comparative analysis of climate change impacts on meteorological, hydrological, and agricultural droughts in the Lake Titicaca basin. Water **13**(2), 175 (2021). DOI 10.3390/w13020175